# Experiment-3

**AIM-** To configure the USART peripheral for asynchronous serial transmission of text data to a terminal.

**APPARATUS-** STM32 Discovery Kit, STM32 CUBE IDE, STM32 CubeMX.

**PROCEDURE**

1. Open CubeMX and Click on ACCESS TO MCU SELECTOR.
2. Write and select the microcontroller as you want to use and then click start the project. Select commercial part number STM32L475VGT6.
3. **Clock Selection**: Go to RCC and configure clock as "By pass Clock Source" as shown below:

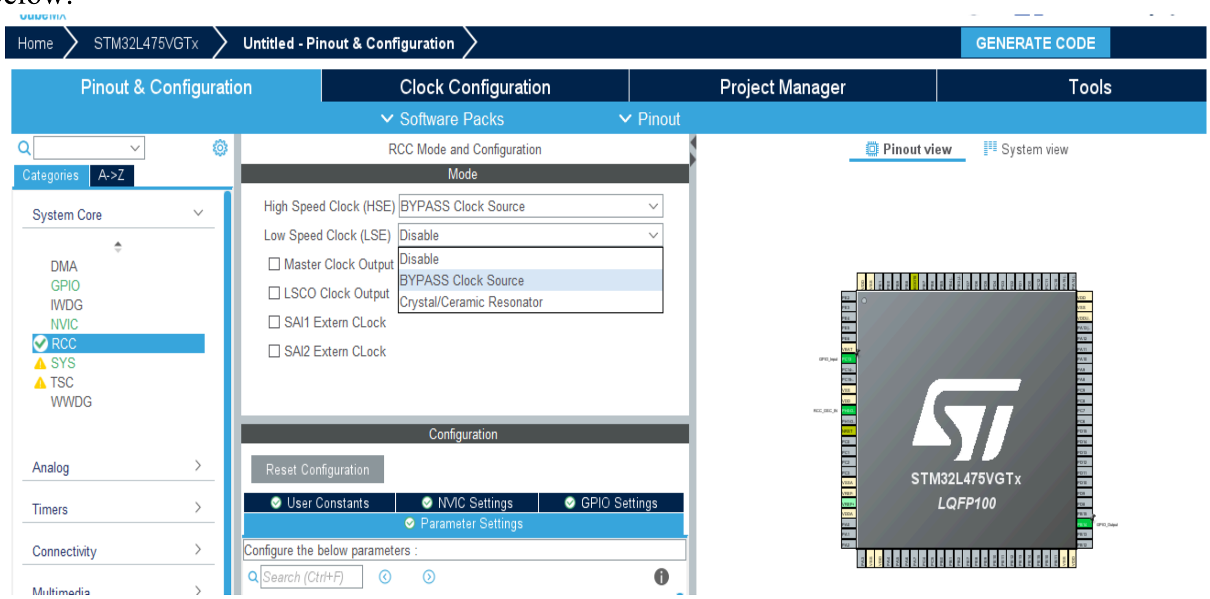


Fig-3.1 – Clock Selection

4. As per user manual of board ST LINK is internally connected to PB6(Tx) and PB7(Rx) of USART1.

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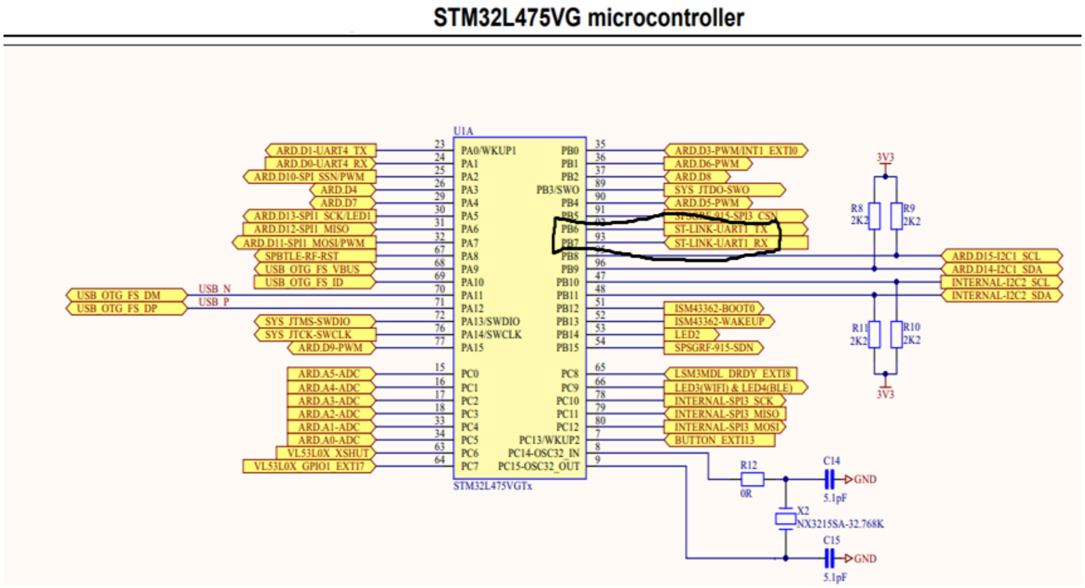


Fig-3.2 – STM32 Board Pin Configuration

5. Under "connectivity" click on " USART" and make PB6 as USART_Tx and PB7 as USART_Rx as shown below:

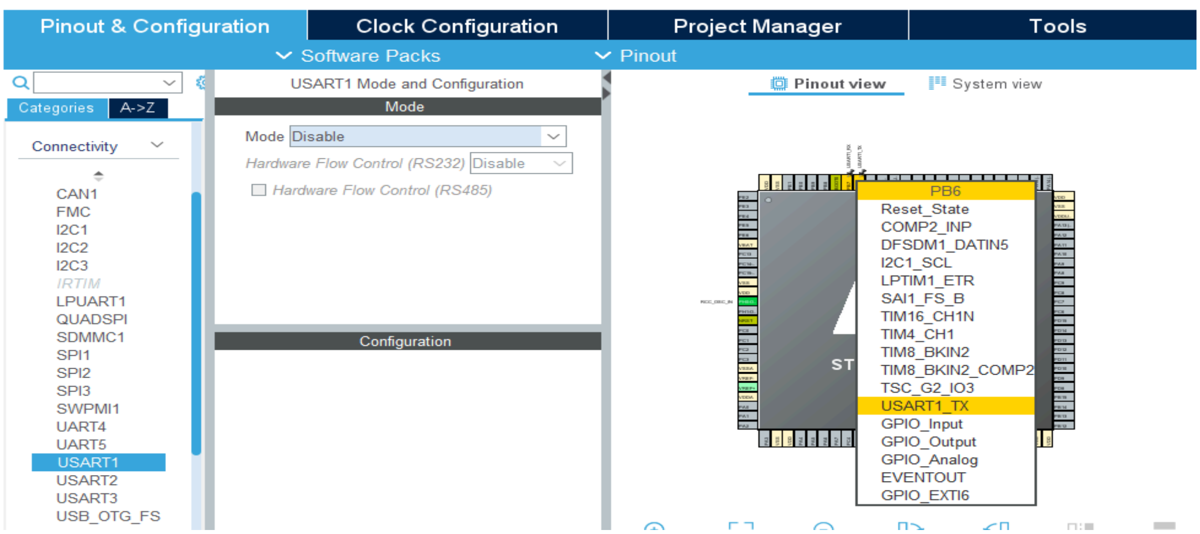


Fig-3.3 – STM32 Board Pinout and Configuration

6. Select USART1 and click on " Asynchronous" in "Mode" as shown below :

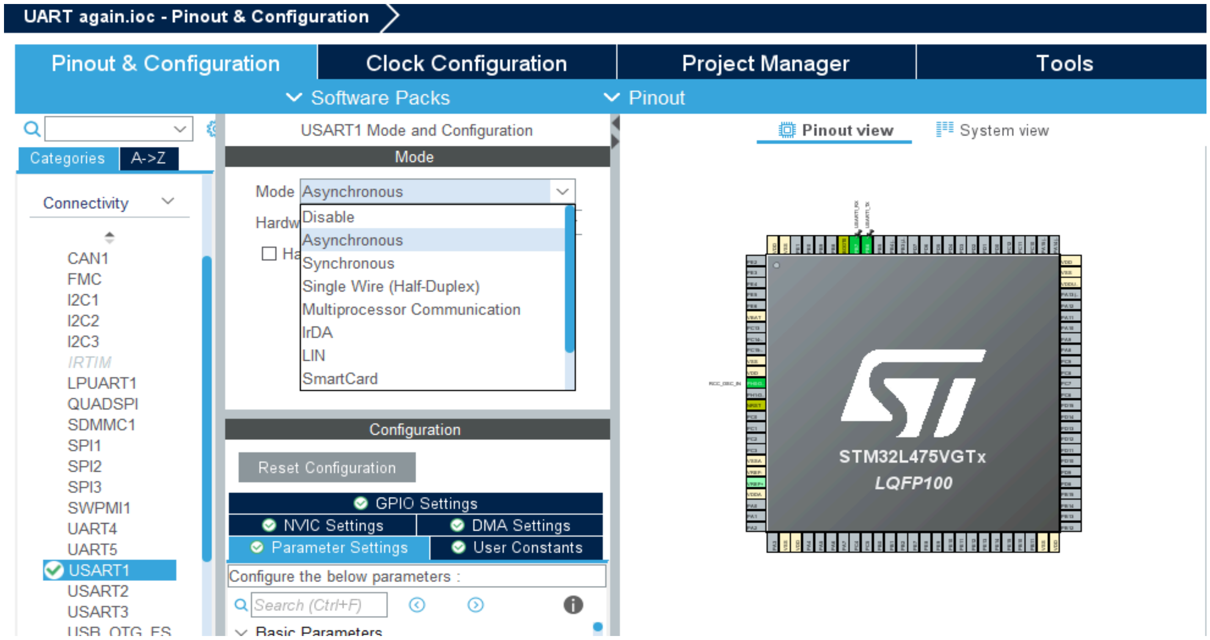


Fig-3.4 – Selection of Asynchronous mode

7. In "parameter setting", the baud rate, word length, stop bits, parity can be changed if required as shown below:

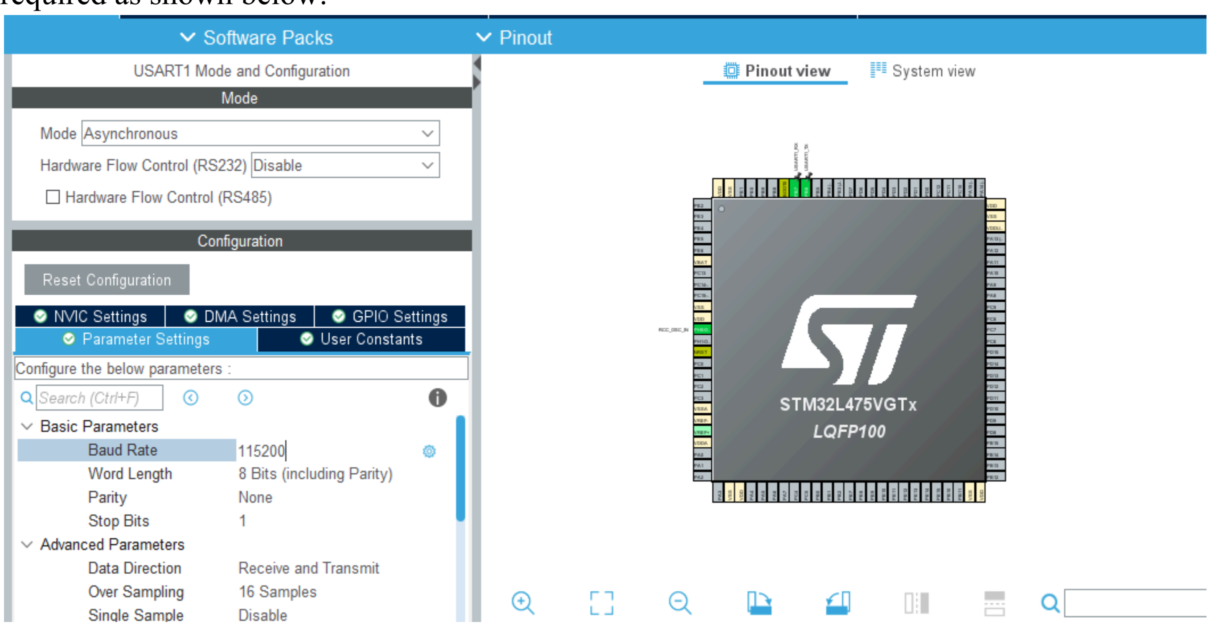


Fig-3.5 – Settings of Basic Parameters

8. **Clock Configuration:** Configure the clock as shown below to get a maximum frequency of 80MHz.

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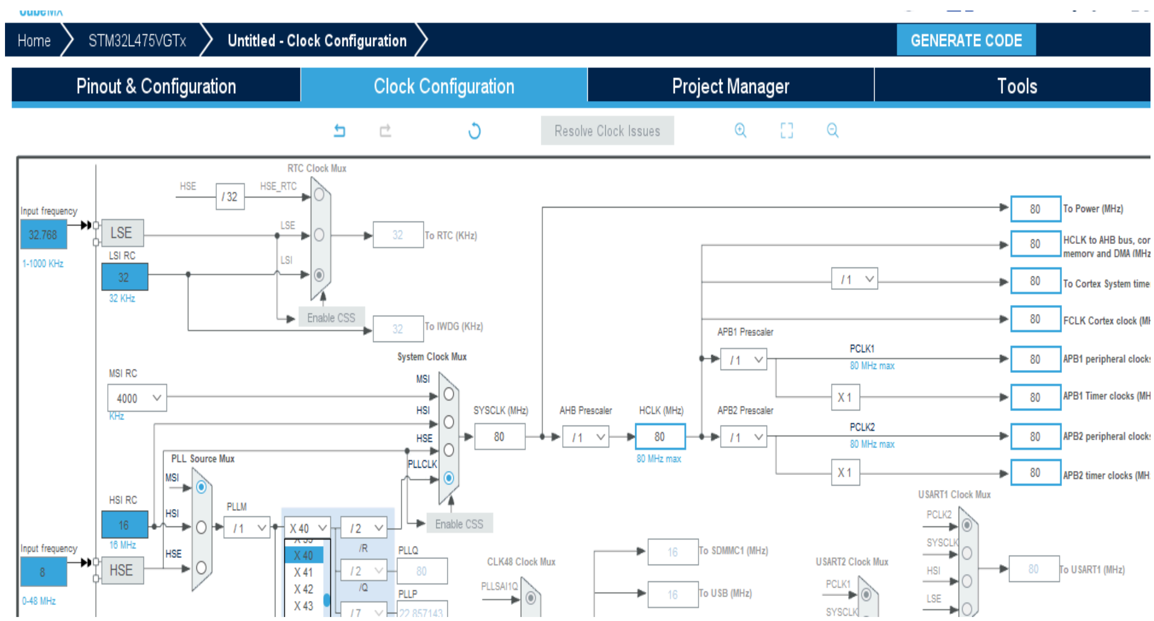


Fig-3.6 – Clock Configuration

9. Write Project name and select IDE as shown below:

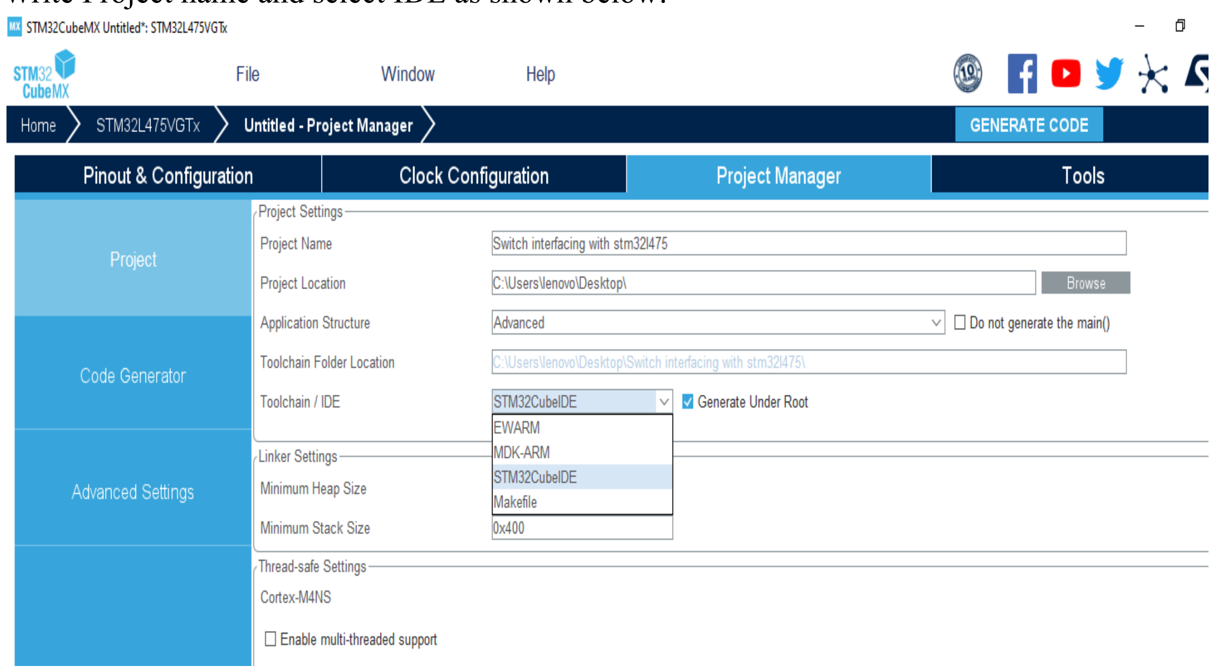


Fig-3.7– Project Name and IDE

10. Click on generate code.

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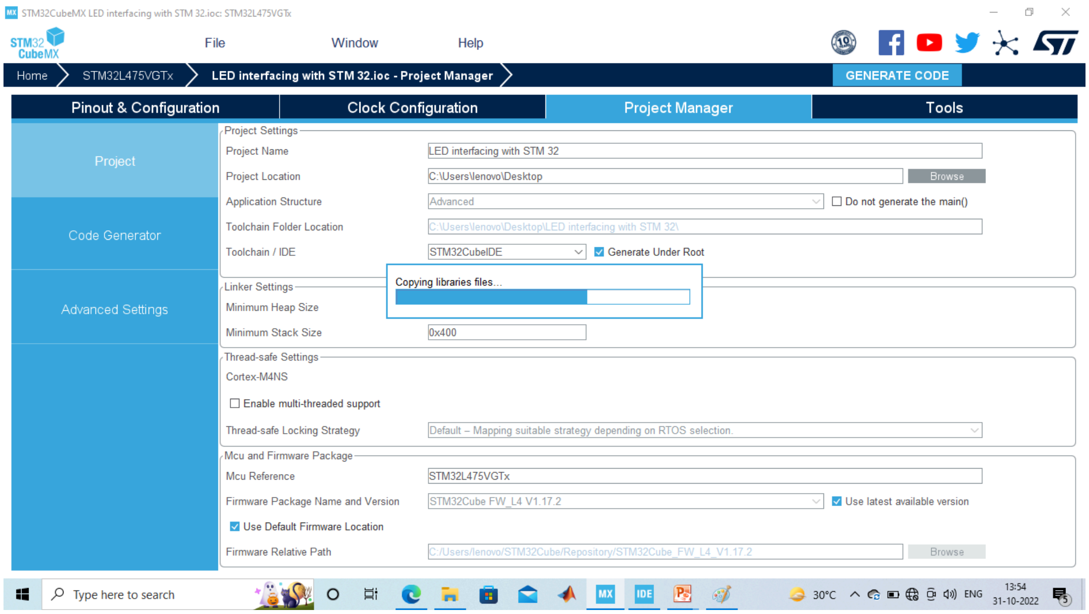


Fig-3.8 – Generate Code

11. Click on open project.
12. Click ‘OK’ on the generated popup showing “successfully imported the project on the work space” and the project will be successfully added to the cube IDE.
13. Open the main source code on Cube IDE by clicking on main.c
14. Go to the main .c and in while(1) loop write following instructions:

*uint8_t Test[] = "Chitkara_University !!!\r\n";*
*HAL_UART_Transmit(&huart1,Test,sizeof(Test),10);*
*HAL_Delay(1000);*

15. Connect “STM32L475VGT6” IoT discovery board with PC or Laptop.
16. Click on “Build” to ensure zero error and click on “Debug” to bring the controller in execution mode as shown below:

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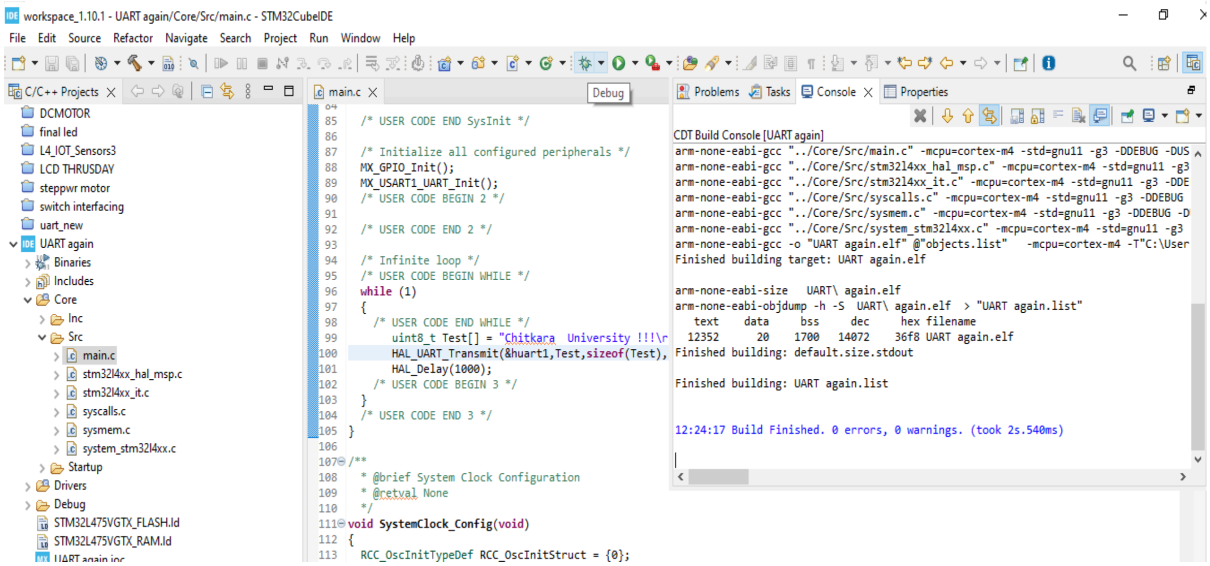

Fig: 3.9 - Program for executing the experiment

17. Click “switch” on popup and then click on “Resume” to execute the program.
18. Open Real term or another console.
19. Select the “Baudrate” same as used in program and select the connected port option.
20. In case of real term after selecting these options click on “change”

**OBSERVATION:** The serial output data can be seen on real time after program execution as shown below:

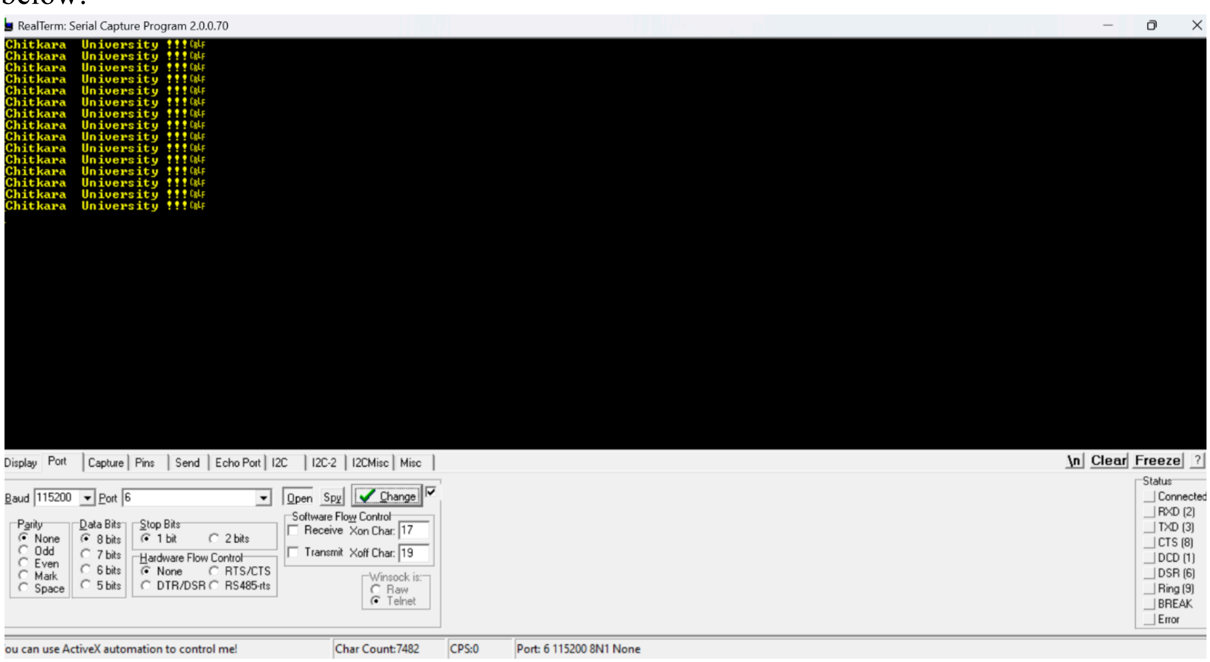

Fig: 3.10 – Output from experiment

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**Reflection summary**

Q1. How does configuring USART in asynchronous mode influence the hardware connections and signals required between the STM32 board and the PC terminal?

Q2. How did the clock configuration (up to 80 MHz) relate to the correct operation of USART in this experiment?

Q3. How did you verify that data was being transmitted periodically (every 1 second) over USART, and what tools or observations helped you confirm this?

Q4.Suppose you want to send sensor data instead of a fixed string. What changes would you make in the code and data handling, based on what you learned in this lab?

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